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SAT 1200+

O'ZBEKISTON DAVLAT UNIVERSITETLARI UCHUN TAYYORLOV

SAT *math* 0-1

FOUNDATION BOOK

QISQA MUDDAT ICHIDA SAT
UCHUN KERAKLI MATEMATIKA
VA IZGLIZ TILI BILIMLARI

1-to'plam. *To'rt bosqichli odatni mustahkamlang: qavs → daraja → II bosqich → I bosqich.*

1. $3 + 4 \times 2 =$

2. $(3 + 4) \times 2 =$

3. $20 - 6 \div 3 =$

4. $(18 - 6) \div 3 =$

5. $2 \times 3^2 =$

6. $(2 \times 3)^2 =$

7. $18 \div 3 \times 2 =$

8. $10 - 2 \times 3 + 4 =$

2-to'plam. *Ishorali sonlar berilgan qiymat sifatida – tartibni qo'llashga e'tibor bering.*

1. $-3 + 4 \times 2 =$

2. $(-3 + 4) \times 2 =$

3. $5 - (-2) \times 3 =$

4. $(-2)^2 \times 3 =$

5. $-2^2 \times 3 =$

6. $10 + (-4) \div 2 =$

7. $(10 + (-4)) \div 2 =$

8. $-5 \times (-3) + 2 =$

3-to'plam. *Kasr chizig'ini guruhlash belgisi sifatida qo'llang.*

1. $\dfrac{3+5}{2\times 2}=$

2. $\dfrac{12-4}{2\times 2}=$

3. $\dfrac{2+3\times 4}{7}=$

4. $\dfrac{5^2-1}{3\times 2}=$

5. $\dfrac{6+2}{2+2}=$

6. $\dfrac{20}{2+3}-1=$

7. $\dfrac{18-6}{3\times(1+1)}=$

8. $\dfrac{7+9}{4^2-12}=$

4-to'plam. *Modul belgisini guruhlash belgisi sifatida qo'llang.*

1. $|5-9|=$

2. $|5-9|+3=$

3. $-|5-9|=$

4. $|3-7|\times 2=$

5. $|-3\times 4|=$

6. $|6-10|-|2-5|=$

7. $10-|4-12|=$

8. $|(-2)^2-9|=$

5-to'plam. *Aralash – haqiqiy test zichligi: daraja, kasr chizig'i va qavs birgalikda.*

1. $\dfrac{2^3+4}{6-4}-1=$

2. $5+\dfrac{(6-2)^2}{4}=$

3. $\dfrac{10-2\times 3}{2^2}=$

4. $3\times(4+2^2)-\dfrac{10}{2}=$

5. $\dfrac{5^2-3\times 5}{2+3}=$

6. $2+3\times(7-2^2)=$

7. $\dfrac{8+|3-7|}{2\times 2}=$

8. $(-3)^2+2\times(5-8)=$

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1. $20\div 4\times 5$ ifodasida ko'rsatilgan yechimda xato bor:
   "1-qadam: $4\times 5=20$. *2-qadam:* $20\div 20=1$."
   Xato va to'g'ri javob: .....

2. $15-3+4$ ifodasida ko'rsatilgan yechimda xato bor:
   "1-qadam: $3+4=7$. *2-qadam:* $15-7=8$."
   Xato va to'g'ri javob: .....

3. $\dfrac{10+2}{2\times 3}$ ifodasida ko'rsatilgan yechimda xato bor:
   "Avval $2\times 3=6$, *so'ng* $2\div 6=\frac{1}{3}$, *so'ng* $10+\frac{1}{3}$."
   Xato va to'g'ri javob: .....

4. -4^2 ifodasida ko'rsatilgan yechimda xato bor:
   "$-4^2=(-4)^2=16$."
   Xato va to'g'ri javob: .....

5. $3-|5-9|$ ifodasida ko'rsatilgan yechimda xato bor:
   "$3-5-9=-11$."
   Xato va to'g'ri javob: .....

7-to’plam. *Bonus / qiyin – har biriga taxminan 90 soniya.*

1. $|(-3)^2 - 2 \times (5 - 8)| \div 3 =$

2. $\dfrac{4 + |2 - 6|^2}{3 \times 2 - 2} =$

3. $2 \times \left[(10 - 4)^2 \div 3 + 1\right] - 5 =$

4. $\dfrac{|-12| + 3^2}{7 - 4} =$

5. $1 + 2 \times \left[|4 - 10| - (3 - 1)^2\right] =$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	11	2	14
	3	18	4	4
	5	18	6	36
	7	12	8	8
2-to'plam	1	5	2	2
	3	11	4	12
	5	-12	6	8
	7	3	8	17
3-to'plam	1	2	2	2
	3	2	4	4
	5	2	6	3
	7	2	8	4
4-to'plam	1	4	2	7
	3	-4	4	8
	5	12	6	1
	7	2	8	5
5-to'plam	1	5	2	9
	3	1	4	19
	5	2	6	11
	7	3	8	3
7-to'plam	1	5	2	5
	3	21	4	7
	5	5	–	–
6-to'plam	1. Xato 1-qadamda – bo'lish chapda, lekin ko'paytirish avval bajarilgan. To'g'ri: $20 \div 4 = 5$, so'ng $5 \times 5 = 25$.			
	2. Xato 1-qadamda – ayirish chapda, lekin qo'shish avval bajarilgan. To'g'ri: $15-3 = 12$, so'ng $12+4 = 16$.			
	3. Kasr chizig'i suratni ham guruhlaydi. To'g'ri: $\frac{12}{6} = 2$.			
	4. Qavs yo'q – daraja faqat 4 ga tegishli. To'g'ri: $-4^2 = -16$.			
	5. Modul ichidagi amal avval bajariladi. To'g'ri: $5 - 9 = -4$, $\lvert -4 \rvert = 4$, $3 - 4 = -1$.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

QR kod

Telegram

1-to'plam. *Qisqartiring – javobni eng sodda ko'rinishda yozing.*

1. $\frac{8}{12} =$
2. $\frac{15}{25} =$
3. $\frac{18}{24} =$
4. $\frac{21}{14} =$
5. $\frac{36}{48} =$
6. $\frac{50}{75} =$
7. $\frac{28}{35} =$
8. $\frac{100}{40} =$

2-to'plam. *Qo'shing yoki ayiring – avval umumiy maxraj toping.*

1. $\frac{1}{3} + \frac{1}{4} =$
2. $\frac{2}{5} + \frac{1}{3} =$
3. $\frac{3}{4} - \frac{1}{6} =$
4. $\frac{5}{6} - \frac{2}{9} =$
5. $\frac{1}{2} + \frac{2}{3} - \frac{1}{4} =$
6. $\frac{7}{10} - \frac{1}{4} =$
7. $\frac{3}{8} + \frac{5}{12} =$
8. $\frac{11}{12} - \frac{3}{4} =$

3-to'plam. *Ko'paytiring – hisoblashdan oldin qisqartirishga harakat qiling.*

1. $\frac{2}{3} \times \frac{3}{5} =$
2. $\frac{4}{7} \times \frac{7}{8} =$
3. $\frac{5}{6} \times \frac{2}{3} =$
4. $\frac{3}{4} \times \frac{8}{9} =$
5. $\frac{7}{10} \times \frac{5}{14} =$
6. $\frac{2}{9} \times \frac{3}{4} =$
7. $\frac{5}{8} \times \frac{4}{15} =$
8. $\frac{9}{10} \times \frac{5}{6} =$

4-to'plam. *Bo'ling – ikkinchi kasrni teskarisiga aylantirib ko'paytiring.*

1. $\frac{2}{3} \div \frac{4}{5} =$
2. $\frac{3}{4} \div \frac{1}{2} =$
3. $\frac{5}{6} \div \frac{5}{9} =$
4. $\frac{7}{8} \div \frac{7}{4} =$
5. $\frac{2}{5} \div \frac{3}{10} =$
6. $\frac{9}{10} \div \frac{3}{5} =$
7. $\frac{4}{9} \div \frac{8}{3} =$
8. $\frac{5}{12} \div \frac{5}{6} =$

5-to'plam. *Aralash son ↔ noto'g'ri kasr.*

1. $2\frac{3}{4}$ ni noto'g'ri kasrga aylantiring.

2. $5\frac{1}{6}$ ni noto'g'ri kasrga aylantiring.

3. $\frac{17}{5}$ ni aralash songa aylantiring.

4. $\frac{29}{4}$ ni aralash songa aylantiring.

5. $1\frac{1}{2} + 2\frac{1}{3} =$ (aralash son)

6. $3\frac{1}{4} - 1\frac{3}{4} =$ (aralash son)

7. $2\frac{2}{5} \times 1\frac{1}{4} =$

8. $\frac{53}{6}$ ni aralash songa aylantiring.

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1. $\frac{1}{3} + \frac{1}{4}$ ifodasida ko'rsatilgan yechimda xato bor:
   "$\frac{1+1}{3+4} = \frac{2}{7}$."
   Xato va to'g'ri javob: .....

2. $\frac{2}{5} + \frac{2}{5}$ ifodasida ko'rsatilgan yechimda xato bor:
   "Suratni saqlab, maxrajlarni qo'shamiz: $\frac{2}{10}$*."*
   Xato va to'g'ri javob: .....

3. $\frac{3}{4} \div \frac{2}{5}$ ifodasida ko'rsatilgan yechimda xato bor:
   "$\frac{4}{3} \times \frac{2}{5} = \frac{8}{15}$."
   Xato va to'g'ri javob: .....

4. $\frac{16}{24}$ ni soddalashtirishda ko'rsatilgan yechimda xato bor:
   "2 ga qisqartirib, $\frac{8}{12}$ *– tayyor."*
   Xato va to'g'ri javob: .....

5. $2\frac{3}{4}$ ni noto'g'ri kasrga aylantirishda ko'rsatilgan yechimda xato bor:
   "$\frac{2+3}{4} = \frac{5}{4}$."
   Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin – har biriga taxminan 90 soniya.*

1. $\left(\frac{1}{2}+\frac{1}{3}\right)\times\frac{2}{5}=$

2. $3\frac{1}{2}\div\left(\frac{1}{4}+\frac{1}{4}\right)=$

3. $\left(\frac{2}{3}-\frac{1}{6}\right)\times\left(\frac{3}{4}+\frac{1}{2}\right)=$

4. $1\frac{3}{4}\times\frac{2}{7}+\frac{1}{8}=$

5. $\left(\frac{5}{6}\div\frac{5}{9}\right)-\left(\frac{1}{2}\times\frac{2}{3}\right)=$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	$\frac{2}{3}$	2	$\frac{3}{5}$
	3	$\frac{3}{4}$	4	$\frac{3}{2}$
	5	$\frac{3}{4}$	6	$\frac{2}{3}$
	7	$\frac{4}{5}$	8	$\frac{5}{2}$
2-to'plam	1	$\frac{7}{12}$	2	$\frac{11}{15}$
	3	$\frac{7}{12}$	4	$\frac{11}{18}$
	5	$\frac{11}{12}$	6	$\frac{9}{20}$
	7	$\frac{19}{24}$	8	$\frac{1}{6}$
3-to'plam	1	$\frac{2}{5}$	2	$\frac{1}{2}$
	3	$\frac{5}{9}$	4	$\frac{2}{3}$
	5	$\frac{1}{4}$	6	$\frac{1}{6}$
	7	$\frac{1}{6}$	8	$\frac{3}{4}$
4-to'plam	1	$\frac{5}{6}$	2	$\frac{3}{2}$
	3	$\frac{3}{2}$	4	$\frac{1}{2}$
	5	$\frac{4}{3}$	6	$\frac{3}{2}$
	7	$\frac{1}{6}$	8	$\frac{1}{2}$
5-to'plam	1	$\frac{11}{4}$	2	$\frac{31}{6}$
	3	$3\frac{2}{5}$	4	$7\frac{1}{4}$
	5	$3\frac{5}{6}$	6	$1\frac{1}{2}$
	7	3	8	$8\frac{5}{6}$
7-to'plam	1	$\frac{1}{3}$	2	7
	3	$\frac{5}{8}$	4	$\frac{5}{8}$
	5	$\frac{7}{6}$	–	–
6-to'plam	1. Surat/maxraj mustaqil qo'shilgan, umumiy maxraj topilmagan. To'g'ri: $\frac{1}{3} = \frac{4}{12}$, $\frac{1}{4} = \frac{3}{12}$, yig'indi $\frac{7}{12}$.			
	2. Maxrajlar teng bo'lsa, faqat suratlar qo'shiladi. To'g'ri: $\frac{2}{5} + \frac{2}{5} = \frac{4}{5}$.			
	3. Teskari kasr ikkinchi (bo'luvchi) kasrga qo'llanishi kerak edi. To'g'ri: $\frac{3}{4} \times \frac{5}{2} = \frac{15}{8}$.			
	4. Bitta qisqartirishdan keyin to'xtagan (8 va 12 hali 4 ga bo'linadi). To'g'ri: $\frac{16}{24} = \frac{2}{3}$.			
	5. Butun son maxrajga ko'paytirilmagan. To'g'ri: $3 \times 4 = 12$, $12 + 3 = 15$, ya'ni $\frac{11}{4}$.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

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1-to’plam. *O’nli kasrlarni qo’shish va ayirish: vergulni tekislashni unutmang.*

1. $3.7 + 2.45 =$

2. $12.6 - 4.75 =$

3. $0.8 + 0.09 =$

4. $15 - 3.25 =$

5. $2.05 + 1.95 =$

6. $100 - 0.4 =$

7. $0.125 + 0.875 =$

8. $7.3 - 2.08 =$

2-to’plam. *O’nli kasrlarni ko’paytirish va bo’lish – kasr xonalarini to’g’ri sanang.*

1. $0.3 \times 0.4 =$

2. $2.5 \times 1.2 =$

3. $0.06 \times 5 =$

4. $1.44 \div 1.2 =$

5. $0.5 \times 0.5 =$

6. $7.2 \div 0.4 =$

7. $0.02 \times 0.03 =$

8. $9.6 \div 2.4 =$

3-to’plam. *O’nli kasr ↔ oddiy kasr almashtirish.*

1. 0.5 ni oddiy kasrga aylantiring.

2. 0.25 ni oddiy kasrga aylantiring.

3. 0.75 ni oddiy kasrga aylantiring.

4. 0.2 ni oddiy kasrga aylantiring.

5. $\frac{3}{4}$ ni o’nli kasrga aylantiring.

6. $\frac{1}{8}$ ni o’nli kasrga aylantiring.

7. $\frac{7}{10}$ ni o’nli kasrga aylantiring.

8. 0.125 ni oddiy kasrga aylantiring.

4-to’plam. *Foiz ↔ o’nli kasr almashtirish – 100% dan katta va 1% dan kichik holatlarni ham o’z ichiga oladi.*

1. 45% ni o’nli kasrga aylantiring.

2. 7% ni o’nli kasrga aylantiring.

3. 150% ni o’nli kasrga aylantiring.

4. 0.5% ni o’nli kasrga aylantiring.

5. 0.6 ni foizga aylantiring.

6. 1.25 ni foizga aylantiring.

7. 0.03 ni foizga aylantiring.

8. 2 ni foizga aylantiring.

5-to'plam. *Foiz ↔ oddiy kasr, va sonning foizini topish.*

1. 80 ning 25% i qanchaga teng?

2. 150 ning 10% i qanchaga teng?

3. 45 ning 60% i qanchaga teng?

4. $\frac{1}{4}$ ni foizga aylantiring.

5. $\frac{3}{5}$ ni foizga aylantiring.

6. 150% ni oddiy kasrga aylantiring.

7. 50 ning 12% i qanchaga teng?

8. 20 ning 90% i qanchaga teng?

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1. 80 ning 45% ini hisoblashda ko'rsatilgan yechimda xato bor:

   "$45 \times 80 = 3600$, *demak javob* 3600."

   Xato va to'g'ri javob: .....

2. $3.7 + 2.45$ ni hisoblashda ko'rsatilgan yechimda xato bor:

   "Vergulni tekislamasdan qo'shib, $3.7 + 2.45 = 2.82$ *deb topilgan."*

   Xato va to'g'ri javob: .....

3. $1.44 \div 1.2$ ni hisoblashda ko'rsatilgan yechimda xato bor:

   "Kasr xonalarini noto'g'ri sanab, $1.44 \div 1.2 = 0.12$ *deb topilgan."*

   Xato va to'g'ri javob: .....

4. 0.2 ni oddiy kasrga aylantirishda ko'rsatilgan yechimda xato bor:

   "0.2 *dagi raqam yuzdan bir xonasida deb o'ylab,* $\frac{2}{100} = \frac{1}{50}$ *deb topilgan."*

   Xato va to'g'ri javob: .....

5. 150% ni o'nli kasrga aylantirishda ko'rsatilgan yechimda xato bor:

   "Vergul faqat bir xona chapga surilib, $150\% = 15$ *deb topilgan."*

   Xato va to'g'ri javob: .....

7-to’plam. *Bonus / qiyin – aralash almashtirishlar, har biriga taxminan 90 soniya.*

1 0.8 ning 35% ini toping, so’ngra natijani 100 ga ko’paytiring.

2 $\frac{3}{8}$ ni foizga aylantiring.

3 0.6 ning necha foizi 0.3 ga teng?

4 1.5 ni foizga va oddiy kasrga aylantiring.

5 (0.25 + 0.5) yig’indisining 40% ini toping.

Javoblar

To’plam	#	Javob	#	Javob
1-to’plam	1	6.15	2	7.85
	3	0.89	4	11.75
	5	4	6	99.6
	7	1	8	5.22
2-to’plam	1	0.12	2	3
	3	0.3	4	1.2
	5	0.25	6	18
	7	0.0006	8	4
3-to’plam	1	1/2	2	1/4
	3	3/4	4	1/5
	5	0.75	6	0.125
	7	0.7	8	1/8
4-to’plam	1	0.45	2	0.07
	3	1.5	4	0.005
	5	60%	6	125%
	7	3%	8	200%
5-to’plam	1	20	2	15
	3	27	4	25%
	5	60%	6	3/2
	7	6	8	18
7-to’plam	1	28	2	37.5%
	3	50%	4	150%, 3/2
	5	0.3	–	–
6-to’plam	1. Foiz to’g’ridan-to’g’ri ko’paytirilgan, o’nli kasrga aylantirilmagan. To’g’ri: $45\% = 0.45$, so’ng $0.45 \times 80 = 36$.			
	2. Vergullar tekislanmagan. To’g’ri: $3.70 + 2.45 = 6.15$.			
	3. Kasr xonalari noto’g’ri sanalgan. To’g’ri: $1.44 \div 1.2 = 1.2$.			
	4. Raqamning xonasi noto’g’ri aniqlangan (yuzdan emas, o’ndan). To’g’ri: $0.2 = \frac{2}{10} = \frac{1}{5}$.			
	5. Vergul bir xona o’rniga ikki xona chapga surilishi kerak edi. To’g’ri: $150\% = 1.50 = 1.5$.			

Video yechimlar va qo’shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

QR kod

Telegram

1-to'plam. *Bir xil ishorali sonlarni qo'shing.*

1. $(-3) + (-8) =$
2. $(-15) + (-9) =$
3. $7 + 12 =$
4. $(-20) + (-5) =$
5. $(-11) + (-14) =$
6. $25 + 16 =$
7. $(-30) + (-12) =$
8. $9 + 18 =$

2-to'plam. *Aralash ishoralar va manfiyni ayirish.*

1. $8 - (-3) =$
2. $(-5) - (-12) =$
3. $(-7) + 10 =$
4. $14 - 20 =$
5. $(-9) - 6 =$
6. $6 - (-6) =$
7. $(-18) + 11 =$
8. $(-4) - (-4) =$

3-to'plam. *Ko'paytirish va bo'lish (ikki ko'paytuvchi).*

1. $(-6) \times 7 =$

2. $(-9) \times (-8) =$

3. $45 \div (-5) =$

4. $(-56) \div (-7) =$

5. $12 \times (-11) =$

6. $(-100) \div 4 =$

7. $(-13) \times (-3) =$

8. $(-81) \div (-9) =$

4-to'plam. *Ko'p ko'paytuvchili zanjirlar – manfiylar sonini sanang.*

1. $(-2) \times (-3) \times (-4) =$

2. $(-5) \times 6 \times (-2) =$

3. $100 \div (-5) \div (-2) =$

4. $(-1) \times (-1) \times (-1) \times (-1) =$

5. $(-3) \times (-3) \times (-3) =$

6. $48 \div (-4) \times (-3) =$

7. $(-10) \times (-10) \div 5 =$

8. $(-2) \times 3 \times (-4) \times (-1) =$

5-to'plam. *Qavslarni oching.*

1. $-(9-4)+6=$

2. $-(5+3)-2=$

3. $10-(6-11)=$

4. $-(-7+2)+1=$

5. $-(8-8)-5=$

6. $3-(-4-6)=$

7. $-(12-20)-9=$

8. $-(-6-(-2))+4=$

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1. $8-(-3)$ ifodasida ko'rsatilgan yechimda xato bor:

   "Noto'g'ri yechim: $8-3=5$*."*

   Xato va to'g'ri javob: .....

2. $(-5)\times(-6)$ ifodasida ko'rsatilgan yechimda xato bor:

   "Noto'g'ri yechim: -30*."*

   Xato va to'g'ri javob: .....

3. $-(10-4)$ ifodasida ko'rsatilgan yechimda xato bor:

   "Noto'g'ri yechim: $-10-4=-14$*."*

   Xato va to'g'ri javob: .....

4. $(-2)\times(-2)\times(-2)$ ifodasida ko'rsatilgan yechimda xato bor:

   "Noto'g'ri yechim: barcha ko'paytuvchilar manfiy bo'lgani uchun javob musbat: 8*."*

   Xato va to'g'ri javob: .....

5. $(-15)\div(-3)$ ifodasida ko'rsatilgan yechimda xato bor:

   "Noto'g'ri yechim: -5*."*

   Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin – har biriga taxminan 90 soniya.*

1. $-(-8+3)\times(-2)-5=$

2. $(-4)\times(-3)-(-7)+(-2)\times 5=$

3. $18\div(-3)+(-2)\times(-4)-6=$

4. $-(5-(-9))+(-14)\div 7=$

5. $(-6)\times[(-2)-3]+(-10)\div(-5)=$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	-11	2	-24
	3	19	4	-25
	5	-25	6	41
	7	-42	8	27
2-to'plam	1	11	2	7
	3	3	4	-6
	5	-15	6	12
	7	-7	8	0
3-to'plam	1	-42	2	72
	3	-9	4	8
	5	-132	6	-25
	7	39	8	9
4-to'plam	1	-24	2	60
	3	10	4	1
	5	-27	6	36
	7	20	8	-24
5-to'plam	1	1	2	-10
	3	15	4	6
	5	-5	6	13
	7	-1	8	8
7-to'plam	1	-15	2	9
	3	-4	4	-16
	5	32	–	–
6-to'plam	1. Minus-minus (manfiyni ayirish) qo'shishga aylantirilmadi. To'g'ri: $8-(-3)=8+3=11$.			
	2. Bir xil ishorali ko'paytirish musbat natija beradi. To'g'ri: $(-5)\times(-6)=30$.			
	3. Manfiy ishora qavs ichidagi faqat birinchi hadga qo'llanildi. To'g'ri: $-(10-4)=-10+4=-6$.			
	4. Uchta (toq sondagi) manfiy ko'paytuvchi bor – toq son manfiylar natijani manfiy qiladi. To'g'ri: $(-2)\times(-2)\times(-2)=-8$.			
	5. Bo'lishda ham bir xil ishora musbat natija beradi. To'g'ri: $(-15)\div(-3)=5$.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

QR kod

Telegram

1-to'plam. *O'xshash hadlarni qo'shing (musbat koeffitsientlar).*

1. $3x + 5x =$

2. $7y + 2y + y =$

3. $4a + 3b + 2a =$

4. $6x + 3 + 2x =$

5. $5m + 4n + 3m + n =$

6. $2x + 3x + 4x =$

7. $8p + 5q + 2p + 3q =$

8. $x + x + x + x =$

2-to'plam. *O'xshash hadlarni qo'shing – ayirish ishtirok etadi.*

1. $7x - 3x =$

2. $5a - 8a =$

3. $4x - 7 + 2x =$

4. $-3y + 5y - 2y =$

5. $6m - 4n - 2m + 7n =$

6. $-5x + 3x - 8 =$

7. $2a - 5a + 6a =$

8. $9x - 4x - x =$

3-to'plam. *Taqsimot qonunini qo'llang (musbat ko'paytuvchi).*

1. $3(x+4) =$
2. $5(2x-3) =$
3. $2(3a+4b) =$
4. $4(x-1) =$
5. $6(2+3x) =$
6. $7(x+y) =$
7. $3(4x-2) =$
8. $2(5x+3) =$

4-to'plam. *Taqsimot qonuni – manfiy ko'paytuvchini har bir hadga tarqating.*

1. $-2(x+3) =$
2. $-(3x-5) =$
3. $-4(2x-1) =$
4. $-(x+7) =$
5. $-3(4-x) =$
6. $-(2x-3y) =$
7. $-5(x+2) =$
8. $-(-x+4) =$

5-to'plam. *Aralash – avval taqsimlang, so'ng o'xshash hadlarni qo'shing.*

1. $3(x+2)+4x=$

2. $5(2x-1)-3x=$

3. $2(x+3)-(x-1)=$

4. $4(x-2)+3(x+1)=$

5. $-(2x+3)+5x=$

6. $3(x-1)-2(x+2)=$

7. $2(3x+1)-(4x-5)=$

8. $5-2(x-3)=$

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri soddalashtirilgan ko'rinishni bering.*

1. $-(3x-5)$ ifodasini soddalashtirishda ko'rsatilgan yechimda xato bor:

   "$-3x-5$"

   Xato va to'g'ri javob: .....

2. $4x+3x^2$ ifodasini soddalashtirishda ko'rsatilgan yechimda xato bor:

   "$7x^2$"

   Xato va to'g'ri javob: .....

3. $-2(x-4)$ ifodasini soddalashtirishda ko'rsatilgan yechimda xato bor:

   "$-2x-8$"

   Xato va to'g'ri javob: .....

4. $5x-(2x-3)$ ifodasini soddalashtirishda ko'rsatilgan yechimda xato bor:

   "$5x-2x-3=3x-3$"

   Xato va to'g'ri javob: .....

5. $3(2x+1)-4$ ifodasini soddalashtirishda ko'rsatilgan yechimda xato bor:

   "$6x+1-4=6x-3$"

   Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin – ichma-ich qavslar va ikki o'zgaruvchili ifodalar.*

1 $2\,[3(x+1)-2x]+5=$

2 $-3(x-2)+2(x+4)-5=$

3 $4x-2\,[x-(3x+1)]=$

4 $5(a-2b)-3(2a-b)=$

5 $-\,[2(x+3)-(x-1)]+4x=$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	$8x$	2	$10y$
	3	$6a + 3b$	4	$8x + 3$
	5	$8m + 5n$	6	$9x$
	7	$10p + 8q$	8	$4x$
2-to'plam	1	$4x$	2	$-3a$
	3	$6x - 7$	4	0
	5	$4m + 3n$	6	$-2x - 8$
	7	$3a$	8	$4x$
3-to'plam	1	$3x + 12$	2	$10x - 15$
	3	$6a + 8b$	4	$4x - 4$
	5	$18x + 12$	6	$7x + 7y$
	7	$12x - 6$	8	$10x + 6$
4-to'plam	1	$-2x - 6$	2	$5 - 3x$
	3	$4 - 8x$	4	$-x - 7$
	5	$3x - 12$	6	$-2x + 3y$
	7	$-5x - 10$	8	$x - 4$
5-to'plam	1	$7x + 6$	2	$7x - 5$
	3	$x + 7$	4	$7x - 5$
	5	$3x - 3$	6	$x - 7$
	7	$2x + 7$	8	$11 - 2x$
7-to'plam	1	$2x + 11$	2	$9 - x$
	3	$8x + 2$	4	$-a - 7b$
	5	$3x - 7$	–	–
6-to'plam	1. Xato – manfiy ishora faqat $3x$ ga tarqatilgan, -5 ga emas. To'g'ri: $-(3x - 5) = 5 - 3x$.			
	2. Xato – x va x^2 o'xshash had emas (daraja boshqa), qo'shib bo'lmaydi. To'g'ri: $3x^2 + 4x$ (allaqachon sodda).			
	3. Xato – $-2 \times (-4) = +8$ bo'lishi kerak edi, ishora tushib qolgan. To'g'ri: $-2(x - 4) = 8 - 2x$.			
	4. Xato – qavs oldidagi manfiy ishora faqat $2x$ ga tarqatilgan, -3 ga emas. To'g'ri: $5x - (2x - 3) = 3x + 3$.			
	5. Xato – 3×1 hisoblanmagan (ikkinchi hadga tarqatish o'tkazib yuborilgan). To'g'ri: $3(2x + 1) - 4 = 6x - 1$.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

QR kod

Telegram

1-to'plam. *Bir qadamli tenglamalar – asosiy amalni teskarisi bilan bajaring.*

1. $x + 7 = 12$

2. $x - 9 = 4$

3. $5x = 35$

4. $\frac{x}{4} = 6$

5. $x - (-3) = 10$

6. $-3x = 21$

7. $x + 15 = 8$

8. $\frac{x}{-5} = -6$

2-to'plam. *Ikki qadamli tenglamalar – qo'shish/ayirishdan so'ng ko'paytirish/bo'lish.*

1. $2x + 3 = 11$

2. $3x - 5 = 10$

3. $\frac{x}{2} + 4 = 9$

4. $5 - 2x = 1$

5. $4x + 7 = -5$

6. $-3x - 2 = 10$

7. $\frac{x}{3} - 1 = -4$

8. $7 - \frac{x}{2} = 3$

3-to'plam. *Qavsni ochish va o'xshash hadlarni birlashtirish.*

1. $3(x+2)=15$
2. $2(x-4)+5=7$
3. $4x+2x-3=15$
4. $-2(x+3)=10$
5. $3(2x-1)-2x=13$
6. $6-3(x-2)=0$
7. $2(3x+1)-(x-5)=22$
8. $-4(x-1)+2x=-2$

4-to'plam. *O'zgaruvchi ikkala tomonda – avval bir tomonga yig'ing.*

1. $3x+5=x+13$
2. $5x-2=2x+7$
3. $4x+1=2x-9$
4. $7-x=2x+1$
5. $2(x+3)=x+10$
6. $3x-4=5(x-2)$
7. $x+8=3x-2$
8. $6-2x=4x-18$

5-to'plam. *Aralash, SAT uslubida – ba'zilari x emas, hosila ifodani so'raydi. Diqqat bilan o'qing.*

1. $2x + 5 = 17$. $x + 3$ ning qiymati nechaga teng?

2. $3(x - 2) = 2x + 1$. $2x$ ning qiymati nechaga teng?

3. $\frac{x}{2} + 3 = 9$. $3x$ ning qiymati nechaga teng?

4. $4x - 7 = x + 8$. $x - 1$ ning qiymati nechaga teng?

5. $2(2x + 1) - 3 = 15$. x ning qiymati nechaga teng?

6. $5x + 2 = 3x + 12$. $5x$ ning qiymati nechaga teng?

7. $6 - x = 2x - 6$. $x + 4$ ning qiymati nechaga teng?

8. $3(x - 1) = x + 5$. $2x - 1$ ning qiymati nechaga teng?

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1. $x - 5 = 12$ ifodasida ko'rsatilgan yechimda xato bor:

   "1-qadam: ikkala tomonga 5 qo'shiladi: $x = 12$."

   Xato va to'g'ri javob: .....

2. $2x - 5 = 9$ ifodasida ko'rsatilgan yechimda xato bor:

   "1-qadam: $2x = 9 - 5 = 4$."

   Xato va to'g'ri javob: .....

3. $-2(x - 3) = 10$ ifodasida ko'rsatilgan yechimda xato bor:

   "1-qadam: $-2x - 6 = 10$."

   Xato va to'g'ri javob: .....

4. $3x + 5 = 20$ ifodasida ko'rsatilgan yechimda xato bor:

   "1-qadam: $8x = 20$, shuning uchun $x = 2.5$."

   Xato va to'g'ri javob: .....

5. $2x+1 = 11$ (savolda $3x$ so'ralgan) ifodasida ko'rsatilgan yechimda xato bor:

   "$x = 5$, javob: 5."

   Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin – kasr va qavslar birgalikda, har biriga taxminan 90 soniya.*

1. $\frac{x+3}{2} = 7$

2. $\frac{3(x-1)}{2} = 6$

3. $2(x+1) - 3(x-2) = 10$

4. $\frac{2x-1}{3} + 1 = 4$

5. $4(x-2) = 2(x+3) - 6$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	5	2	13
	3	7	4	24
	5	7	6	-7
	7	-7	8	30
2-to'plam	1	4	2	5
	3	10	4	2
	5	-3	6	-4
	7	-9	8	8
3-to'plam	1	3	2	5
	3	3	4	-8
	5	4	6	4
	7	3	8	3
4-to'plam	1	4	2	3
	3	-5	4	2
	5	4	6	3
	7	5	8	4
5-to'plam	1	9	2	14
	3	36	4	4
	5	4	6	25
	7	8	8	7
7-to'plam	1	11	2	5
	3	-2	4	5
	5	4	–	–
6-to'plam	1. 5 faqat o'ng tomonga qo'shilgan – ikkala tomonga qo'shish kerak. To'g'ri: $x = 12 + 5 = 17$.			
	2. -5 ishorasi almashtirilmagan. To'g'ri: $2x = 9 + 5 = 14$, $x = 7$.			
	3. -2 faqat x ga taqsimlangan, -3 ga emas. To'g'ri: $-2x + 6 = 10 \Rightarrow x = -2$.			
	4. $3x$ va 5 o'xshash had emas. To'g'ri: $3x = 15 \Rightarrow x = 5$.			
	5. $x = 5$ to'g'ri, lekin savol $3x$ ni so'ragan. To'g'ri: $3x = 15$.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

QR kod

Telegram

1-to'plam. *Daraja qoidalari – nol/manfiy ko'rsatkichsiz sof mashq.*

1. $2^2 \times 2^3 =$

2. $\frac{3^4}{3^2} =$

3. $(2^2)^3 =$

4. $5^3 \times 5^2 =$

5. $\frac{4^5}{4^3} =$

6. $(3^2)^2 =$

7. $2^3 \times 2^3 =$

8. $(2^3)^3 =$

2-to'plam. *Nol va manfiy ko'rsatkichlar.*

1. $5^0 =$

2. $2^{-3} =$

3. $4^{-2} =$

4. $2^0 \times 2^{-2} =$

5. $3^{-2} \times 3^4 =$

6. $5^2 \times 5^{-2} =$

7. $10^{-1} \times 100 =$

8. $2^{-1} + 2^{-2} =$

3-to'plam. *Aralash – haqiqiy test zichligi.*

1. $\dfrac{(2^2 \times 2^3)^2}{2^4} =$

2. $(3 \times 3^2)^2 =$

3. $\dfrac{5^3}{5^2 \times 5^{-1}} =$

4. $(2^{-1})^{-3} =$

5. $(4^2)^0 =$

6. $\dfrac{2^5}{2^{-2}} =$

7. $\dfrac{(3^2)^2}{3^3} =$

8. $10^3 \times 10^{-1} \div 10 =$

4-to'plam. *Kvadrat ildizlarni soddalashtirish.*

1. $\sqrt{16} =$

2. $\sqrt{49} =$

3. $\sqrt{18} =$

4. $\sqrt{50} =$

5. $\sqrt{72} =$

6. $\sqrt{20} =$

7. $\sqrt{45} =$

8. $\sqrt{8} =$

5-to'plam. *Ildizlarni qo'shish va ayirish.*

1. $\sqrt{8} + \sqrt{2} =$

2. $\sqrt{18} - \sqrt{2} =$

3. $\sqrt{12} + \sqrt{27} =$

4. $\sqrt{50} - \sqrt{8} =$

5. $2\sqrt{3} \times \sqrt{3} =$

6. $\sqrt{2} \times \sqrt{8} =$

7. $3\sqrt{5} + 2\sqrt{5} =$

8. $\sqrt{45} - \sqrt{20} =$

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1. $2^3 \times 2^3$ ifodasida ko'rsatilgan yechimda xato bor:
   "$2^3 \times 2^3 = 2^9 = 512$."
   Xato va to'g'ri javob: .....

2. $(2^2)^3$ ifodasida ko'rsatilgan yechimda xato bor:
   "$(2^2)^3 = 2^{2+3} = 2^5 = 32$."
   Xato va to'g'ri javob: .....

3. 2^{-3} ifodasida ko'rsatilgan yechimda xato bor:
   "$2^{-3} = -8$."
   Xato va to'g'ri javob: .....

4. $\sqrt{16 + 9}$ ifodasida ko'rsatilgan yechimda xato bor:
   "$\sqrt{16} + \sqrt{9} = 4 + 3 = 7$."
   Xato va to'g'ri javob: .....

5. $\sqrt{32}$ ifodasida ko'rsatilgan yechimda xato bor:
   "$32 = 16 \times 2$ *deb to'g'ri ajratilgan, lekin javobda* $16\sqrt{2}$ *deb yozilgan.*"
   Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin – har biriga taxminan 90 soniya.*

1. $\dfrac{(2^3)^2}{2^4} + \sqrt{16} =$

2. $\sqrt{3^2 \times 2^2} =$

3. $2^{-2} \times 4^2 + \sqrt{9} =$

4. $(\sqrt{2})^4 =$

5. $1 + \sqrt{5^2 - 3^2} =$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	32	2	9
	3	64	4	3125
	5	16	6	81
	7	64	8	512
2-to'plam	1	1	2	$\frac{1}{8}$
	3	$\frac{1}{16}$	4	$\frac{1}{4}$
	5	9	6	1
	7	10	8	$\frac{3}{4}$
3-to'plam	1	64	2	729
	3	25	4	8
	5	1	6	128
	7	3	8	10
4-to'plam	1	4	2	7
	3	$3\sqrt{2}$	4	$5\sqrt{2}$
	5	$6\sqrt{2}$	6	$2\sqrt{5}$
	7	$3\sqrt{5}$	8	$2\sqrt{2}$
5-to'plam	1	$3\sqrt{2}$	2	$2\sqrt{2}$
	3	$5\sqrt{3}$	4	$3\sqrt{2}$
	5	6	6	4
	7	$5\sqrt{5}$	8	$\sqrt{5}$
7-to'plam	1	8	2	6
	3	7	4	4
	5	5	–	–
6-to'plam	1. Ko'rsatkichlarni ko'paytirgan, qo'shish kerak edi (ko'paytma qoidasi). To'g'ri: $2^{3+3} = 2^6 = 64$.			
	2. Ko'rsatkichlarni qo'shgan, ko'paytirish kerak edi (daraja darajasi). To'g'ri: $2^{2\times3} = 2^6 = 64$.			
	3. Manfiy ko'rsatkichni asosni manfiy qiladi deb tushungan. To'g'ri: $2^{-3} = \frac{1}{8}$.			
	4. Yig'indining ildizini ildizlar yig'indisiga bo'lgan – bu hech qachon to'g'ri emas. To'g'ri: $\sqrt{25} = 5$.			
	5. 16 ni to'g'ri ajratgan, lekin uning ildizini olmagan. To'g'ri: $\sqrt{16 \times 2} = 4\sqrt{2}$.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

QR kod

Telegram

1-to'plam. *Nuqta qaysi chorakka tegishli ekanini aniqlang.*

1. $(3,\ 5)$ → chorak:

2. $(-2,\ 4)$ → chorak:

3. $(-6,\ -1)$ → chorak:

4. $(7,\ -3)$ → chorak:

5. $(-4,\ -8)$ → chorak:

6. $(5,\ -5)$ → chorak:

7. $(-9,\ 2)$ → chorak:

8. $(1,\ 1)$ → chorak:

2-to'plam. *Tenglamadan k (burchak koeffitsienti) va b ni ajrating.*

1. $y = 3x + 7$ → $k=$ $b=$

2. $y = -2x + 1$ → $k=$ $b=$

3. $y = x - 5$ → $k=$ $b=$

4. $y = -x + 9$ → $k=$ $b=$

5. $y = 4x$ → $k=$ $b=$

6. $y = -6$ → $k=$ $b=$

7. $y = 0.5x - 3$ → $k=$ $b=$

8. $y = \frac{1}{3}x - 2$ → $k=$ $b=$

3-to'plam. *$f(a)$ ni musbat qiymatlar uchun hisoblang.*

1. $f(x) = 2x + 3.\ f(4) =$

2. $f(x) = 5x - 1.\ f(2) =$

3. $f(x) = -3x + 4.\ f(1) =$

4. $f(x) = x + 10.\ f(0) =$

5. $f(x) = 4x.\ f(3) =$

6. $f(x) = -x + 6.\ f(5) =$

7. $f(x) = 2x - 7.\ f(10) =$

8. $f(x) = -5x + 2.\ f(3) =$

4-to'plam. *$f(a)$ ni manfiy qiymatlar uchun hisoblang – manfiy son har doim qavsda.*

1. $f(x) = 3x + 2.\ f(-2) =$

2. $f(x) = -4x + 1.\ f(-3) =$

3. $f(x) = x + 8.\ f(-8) =$

4. $f(x) = x^2.\ f(-3) =$

5. $f(x) = x^2 + 1.\ f(-4) =$

6. $f(x) = 2x^2 - 3.\ f(-2) =$

7. $f(x) = -x^2.\ f(-5) =$

8. $f(x) = (x + 1)^2.\ f(-6) =$

5-to'plam. *Aralash – shaklni almashtiring, qiymatni hisoblang, tenglamani yeching.*

1 $y = 5x - 3$ ni $f(x)$ shaklida yozing va $f(2)$ ni toping.

2 $f(x) = -2x + 8$ ni $y = kx + b$ shaklida yozing, k, b ni toping.

3 $y = x + 1$ ni $f(x)$ shaklida yozing, so'ng $f(-4)$ ni hisoblang.

4 $f(x) = 4x - 9$. Agar $f(x) = 11$ bo'lsa, x ni toping.

5 $f(x) = -3x + 6$. Agar $f(x) = 0$ bo'lsa, x ni toping.

6 $y = -x - 5$ ni $f(x)$ shaklida yozing va $f(0)$ ni toping.

7 $f(x) = 2x + 1$ bo'lsa, $f(3) + f(-3)$ ni hisoblang.

8 $f(x) = x^2 - 4$. Agar $f(a) = 5$ bo'lsa, a ning musbat qiymatini toping.

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1 $f(x) = 3x + 2$. $f(5)$ ni toping:

"$f(5) = f \cdot 5$, f noma'lum, javob topib bo'lmaydi."

Xato va to'g'ri javob: .....

2 $f(x) = x^2$. $f(-4)$ ni toping:

"$f(-4) = -4^2 = -16$."

Xato va to'g'ri javob: .....

3 $(-3,\ 5)$ nuqtasi qaysi chorakda?

"$x = -3$ ni musbat deb o'ylab, I chorak."

Xato va to'g'ri javob: .....

4 $f(x) = 2x - 5$. Agar $f(x) = 7$, x ni toping:

"$2x - 5 = 7 \Rightarrow 2x = 2 \Rightarrow x = 1$."

Xato va to'g'ri javob: .....

5 $f(x) = x^2 + 3$. $f(-2)$ ni toping:

"$f(-2) = -2^2 + 3 = -4 + 3 = -1$."

Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin – har biriga taxminan 90 soniya.*

1 $f(x) = 2x^2 - 3x + 1$. $f(-2) =$

2 $f(x) = -(x-3)^2$. $f(6) =$

3 $y = -4x + 11$ ni $f(x)$ shaklida yozing, so'ng $f(x) = 3$ bo'lganda x ni toping.

4 $f(x) = x^2 - 2x$. Agar $f(a) = 8$ bo'lsa, a ning manfiy qiymatini toping.

5 $f(x) = 3x + c$. Agar $f(2) = 10$ bo'lsa, c ni toping.

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	I	2	II
	3	III	4	IV
	5	III	6	IV
	7	II	8	I
2-to'plam	1	k=3, b=7	2	k=−2, b=1
	3	k=1, b=−5	4	k=−1, b=9
	5	k=4, b=0	6	k=0, b=−6
	7	k=0.5, b=−3	8	$k=\frac{1}{3}$, b=−2
3-to'plam	1	11	2	9
	3	1	4	10
	5	12	6	1
	7	13	8	−13
4-to'plam	1	−4	2	13
	3	0	4	9
	5	17	6	5
	7	−25	8	25
5-to'plam	1. $f(x) = 5x-3$; $f(2) = 7$ 2. $y = -2x+8$; $k = -2$, $b = 8$			
	3. $f(x) = x + 1$; $f(-4) = -3$ 4. $x = 5$			
	5. $x = 2$ 6. $f(x) = -x - 5$; $f(0) = -5$			
	7. $f(3) + f(-3) = 7 + (-5) = 2$			
	8. $a = 3$			
7-to'plam	1	15	2	−9
	3. $f(x) = -4x + 11$; $x = 2$ 4. $a = -2$ 5. $c = 4$			
6-to'plam	1. f funksiya nomi, songa ko'paytirilmaydi. To'g'ri: $f(5) = 3(5) + 2 = 17$.			
	2. Manfiy kiritmada qavs tushib qolgan. To'g'ri: $f(-4) = (-4)^2 = 16$.			
	3. x ning ishorasi noto'g'ri o'qilgan. To'g'ri: x= −3<0, y=5>0 ⇒ II chorak.			
	4. 7 + 5 = 12, 2 emas. To'g'ri: $2x = 12 \Rightarrow x = 6$.			
	5. Yana qavs tushib qolgan. To'g'ri: $f(-2) = (-2)^2+3 = 7$.			

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1-to'plam. *Amallar tartibi va kasrlarni birlashtiring.*

1. $\frac{3}{4} + 2 \times \left(\frac{1}{2} - \frac{1}{8}\right) =$

2. $\left(\frac{2}{3} + \frac{1}{6}\right) \times 4 - 1 =$

3. $10 - 3 \times \left(\frac{2}{5} + \frac{1}{10}\right) =$

4. $\left(\frac{7}{8} - \frac{1}{4}\right) \div \frac{1}{2} =$

5. $2 \times \frac{3}{4} + \left(1 - \frac{1}{3}\right) \times 6 =$

6. $\left(\frac{1}{2} + \frac{1}{3} + \frac{1}{6}\right) \times 12 - 5 =$

2-to'plam. *Ishorali sonlar va darajalarni birlashtiring.*

1. $(-2)^3 - 3 \times (-4) =$

2. $-3^2 + (-3)^2 =$

3. $(-5)^2 \div (-5) + 2^3 =$

4. $-2 \times (-3)^2 + (-4) =$

5. $(-1)^5 \times (-2)^3 =$

6. $4 - (-2)^4 \div 2 =$

3-to'plam. *O'nli kasr/foiz va tenglamalarni birlashtiring. x ni toping.*

1. $0.25x + 10 = 35$

2. 40% of x is 32

3. $x - 15\%$ of $x = 68$

4. $0.5(x + 4) = 13$

5. 20% of $(x + 10) = 16$

6. $x + 0.1x = 44$

4-to'plam. *Algebraik ifodalar va tenglamalarni birlashtiring. x ni toping.*

1. $3(x - 2) + 5 = 2x + 7$

2. $2(x + 3) - (x - 1) = 10$

3. $4x - (2x + 5) = 7$

4. $5(x - 1) = 3(x + 3)$

5. $-2(x + 4) + 6 = x - 2$

6. $3(2x - 1) - 4 = 2(x + 5)$

5-to'plam. *Kasrli tenglamalarni yeching. x ni toping.*

1. $\frac{2}{3}x - 1 = 5$

2. $\frac{x}{4} + 3 = 7$

3. $\frac{3}{4}x + \frac{1}{2} = 2$

4. $\frac{x+2}{3} = 5$

5. $\frac{2x-1}{5} = 3$

6. $\frac{x}{2} - \frac{x}{3} = 2$

6-to'plam. *$f(x)$ funksiyasi, ishorali sonlar va darajalar.*

1. $f(x) = -2x^2 + 3$. $f(-2) =$

2. $f(x) = x^3 - 4x$. $f(-1) =$

3. $f(x) = (x-2)^2$. $f(-3) =$

4. $f(x) = 2x^2 - 5x + 1$. $f(-1) =$

5. $f(x) = -x^2 + 2x$. $f(3) =$? $f(-3) =$

6. $f(x) = kx + b$. $f(2) = 7, f(0) = 3$. $k =$? $b =$

7-to'plam. *Xatoni toping – ikki mavzu aralashgan, bitta xato qadam bor.*

1 $\frac{1}{2}x + 3 = 7$ ifodasida ko'rsatilgan yechimda xato bor:

"$\frac{1}{2}x = 7 + 3 = 10$, *shuning uchun* $x = 20$."

Xato va to'g'ri javob: .....

2 $3(x - 2) = 9$ ifodasida ko'rsatilgan yechimda xato bor:

"$3x - 2 = 9$, *shuning uchun* $3x = 11$, $x = \frac{11}{3}$."

Xato va to'g'ri javob: .....

3 $f(x) = x^2 - 2x$, $f(-3)$ ifodasida ko'rsatilgan yechimda xato bor:

"$f(-3) = (-3)^2 - 2(-3) = 9 - 6 = 3$."

Xato va to'g'ri javob: .....

4 $0.4x = 24$ ifodasida ko'rsatilgan yechimda xato bor:

"$x = 24 \times 0.4 = 9.6$."

Xato va to'g'ri javob: .....

5 $2(x + 1) - x = 9$ ifodasida ko'rsatilgan yechimda xato bor:

"$2x + 1 - x = 9$, $x + 1 = 9$, $x = 8$."

Xato va to'g'ri javob: .....

8-to'plam. *Bonus / eng qiyin – bitta masalada 3+ ko'nikma.*

1 $\frac{1}{2}(x - 4) + 0.25x = 5$

2 $f(x) = -2(x - 1)^2 + 3$. $f(-1) =$

3 30% of $(2x + 10) = x + 2$

4 $\frac{3}{4}x - (-2)^3 = 11$

5 $f(x) = kx + b$. $f(-2) = 1$, $f(2) = 9$. $k =$? $b =$? $f(0) =$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	$\frac{3}{2}$	2	$\frac{7}{3}$
	3	$\frac{17}{2}$	4	$\frac{5}{4}$
	5	$\frac{11}{2}$	6	7
2-to'plam	1	4	2	0
	3	3	4	-22
	5	8	6	-4
3-to'plam	1	$x=100$	2	$x=80$
	3	$x=80$	4	$x=22$
	5	$x=70$	6	$x=40$
4-to'plam	1	$x=8$	2	$x=3$
	3	$x=6$	4	$x=7$
	5	$x=0$	6	$x=\frac{17}{4}$
5-to'plam	1	$x=9$	2	$x=16$
	3	$x=2$	4	$x=13$
	5	$x=8$	6	$x=12$
6-to'plam	1	-5	2	3
	3	25	4	8
	5	$f(3)=-3$, $f(-3)=-15$		
	6	$k=2$, $b=3$		
7-to'plam	1. Teskari amal xato – 3 qo'shish o'rniga ayirish kerak edi. To'g'ri: $\frac{1}{2}x = 4$, $x = 8$.			
	2. 3 faqat x ga taqsimlangan, -2 ga emas. To'g'ri: $3x - 6 = 9$, $x = 5$.			
	3. $-2 \times (-3) = -6$ emas, $+6$. To'g'ri: $9 + 6 = 15$.			
	4. Ko'paytirish o'rniga bo'lish kerak edi. To'g'ri: $x = 24 \div 0.4 = 60$.			
	5. 2 faqat x ga taqsimlangan, $+1$ ga emas. To'g'ri: $2x + 2 - x = 9$, $x = 7$.			
8-to'plam	1	$x=\frac{28}{3}$		
	2	-5	3	$x=\frac{5}{2}$
	4	$x=4$	5	$k=2, b=5$

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